

# DevOps CheatSheet



git



kubernetes



docker



Terraform

# AWS Cheat Sheet

## Installation

See <https://docs.aws.amazon.com/cli/latest/userguide/getting-started-install.html>

For autocompletion see <https://docs.aws.amazon.com/cli/latest/userguide/cli-configure-completion.html>.

## Global flags

| Flag                 | Description                               |
|----------------------|-------------------------------------------|
| --profile <name>     | Run a command using the specified profile |
| --output <style>     | Format output as (json/text/table/yaml)   |
| --help               | Show help                                 |
| --endpoint-url <url> | Override default API endpoint             |
| --region             | Override default region                   |
| --version            | Show version of AWS CLI                   |

## Configuration

| Command                         | Description                                   |
|---------------------------------|-----------------------------------------------|
| aws configure                   | Configure AWS                                 |
| aws configure sso               | Configure AWS IAM Identity Center credentials |
| aws configure list              | Show current configuration data               |
| aws configure set <key> <value> | Set a configuration value                     |
| aws configure get <key>         | Show a configuration value                    |

## Profiles

| Command                        | Description             |
|--------------------------------|-------------------------|
| aws configure --profile <name> | Create a new profile    |
| aws configure list-profiles    | Show available profiles |

## Account information

| Command                     | Description                  |
|-----------------------------|------------------------------|
| aws sts get-caller-identity | Retrieve account information |

## Identity and Access Management (IAM)

| Command                                                                       | Description     |
|-------------------------------------------------------------------------------|-----------------|
| aws iam list-users                                                            | List users      |
| aws iam list-groups                                                           | List groups     |
| aws iam list-roles                                                            | List roles      |
| aws iam list-policies                                                         | List policies   |
| aws iam create-user --user-name <name>                                        | Create a user   |
| aws iam create-group --group-name <name>                                      | Create a group  |
| aws iam create-role --role-name <name> --assume-role-policy-document <policy> | Create a role   |
| aws iam create-policy --policy-name <name> --policy-document <policy>         | Create a policy |

## Elastic Compute Cloud (EC2)

| Command                                                      | Description           |
|--------------------------------------------------------------|-----------------------|
| aws ec2 describe-instances                                   | List instances        |
| aws ec2 run-instances --image-id <id> --instance-type <type> | Create an instance    |
| aws ec2 terminate-instances --instance-ids <id>              | Terminate an instance |
| aws ec2 start-instances --instance-ids <id>                  | Start an instance     |
| aws ec2 stop-instances --instance-ids <id>                   | Stop an instance      |

## Elastic Kubernetes Service (EKS)

| Command                                | Description        |
|----------------------------------------|--------------------|
| aws eks list-clusters                  | List clusters      |
| aws eks describe-cluster --name <name> | Describe a cluster |
| aws eks create-cluster --name <name>   | Create a cluster   |
| aws eks delete-cluster --name <name>   | Delete a cluster   |

## Relational Database Service (RDS)

| Command                            | Description                    |
|------------------------------------|--------------------------------|
| aws rds describe-db-clusters       | List clusters                  |
| aws rds describe-db-instances      | List instances                 |
| aws rds create-db-cluster-snapshot | Create a snapshot of a cluster |

## Simple Storage Service (S3)

| Command                          | Description                            |
|----------------------------------|----------------------------------------|
| aws s3 ls                        | List buckets                           |
| aws s3 ls s3://<bucket>          | List objects in a bucket               |
| aws s3 mb s3://<bucket>          | Create a bucket                        |
| aws s3 rb s3://<bucket>          | Delete a bucket                        |
| aws s3 cp <src> <dest>           | Copy a file from/to S3                 |
| aws s3 rm s3://<bucket>/<object> | Delete an object                       |
| aws s3 sync s3://bucket <dir>    | Sync a bucket to a local directory     |
| aws s3 sync <dir> s3://bucket    | Sync a local directory to an S3 bucket |

## Other commands

| Command            | Description                                |
|--------------------|--------------------------------------------|
| aws --version      | Show version                               |
| aws help           | Show help including all available commands |
| aws <command> help | Show help for a specific command           |

# Docker Cheat Sheet



## Installation

Install Docker Engine (<https://docs.docker.com/engine/install/>, Linux only) or Docker Desktop (<https://docs.docker.com/desktop/>, Linux, macOS and Windows).

## Container commands

| Command                                                                 | Description                                 |
|-------------------------------------------------------------------------|---------------------------------------------|
| <code>docker run &lt;image&gt;</code>                                   | Create and run a new container              |
| <code>docker run -p 8080:80 &lt;image&gt;</code>                        | Publish container port 80 to host port 8080 |
| <code>docker run -d &lt;image&gt;</code>                                | Run a container in the background           |
| <code>docker run -v &lt;host&gt;:&lt;container&gt; &lt;image&gt;</code> | Mount a host directory to a container       |
| <code>docker ps</code>                                                  | List currently running containers           |
| <code>docker ps --all</code>                                            | List all containers (running or stopped)    |
| <code>docker logs &lt;container_name&gt;</code>                         | Fetch the logs of a container               |
| <code>docker logs -f &lt;container_name&gt;</code>                      | Fetch and follow the logs of a container    |
| <code>docker stop &lt;container_name&gt;</code>                         | Stop a running container                    |
| <code>docker start &lt;container_name&gt;</code>                        | Start a stopped container                   |
| <code>docker rm &lt;container_name&gt;</code>                           | Remove a container                          |

## Executing commands in a container

| Command                                                         | Description                              |
|-----------------------------------------------------------------|------------------------------------------|
| <code>docker exec &lt;container_name&gt; &lt;command&gt;</code> | Execute a command in a running container |
| <code>docker exec -it &lt;container_name&gt; bash</code>        | Open a shell in a running container      |

## Image commands

| Command                                      | Description                                                               |
|----------------------------------------------|---------------------------------------------------------------------------|
| <code>docker build -t &lt;image&gt; .</code> | Build a new image from the Dockerfile in the current directory and tag it |
| <code>docker images</code>                   | List local images                                                         |
| <code>docker rmi &lt;image&gt;</code>        | Remove an image                                                           |

## Container registry commands

| Command                                   | Description                          |
|-------------------------------------------|--------------------------------------|
| <code>docker login</code>                 | Login to Docker Hub                  |
| <code>docker login &lt;server&gt;</code>  | Login to another container registry  |
| <code>docker logout</code>                | Logout of Docker Hub                 |
| <code>docker logout &lt;server&gt;</code> | Logout of another container registry |
| <code>docker push &lt;image&gt;</code>    | Upload an image to a registry        |
| <code>docker pull &lt;image&gt;</code>    | Download an image from a registry    |
| <code>docker search &lt;image&gt;</code>  | Search Docker Hub for images         |

## System commands

| Command                             | Description            |
|-------------------------------------|------------------------|
| <code>docker system df</code>       | Show Docker disk usage |
| <code>docker system prune</code>    | Remove unused data     |
| <code>docker system prune -a</code> | Remove all unused data |

## Docker Compose

| Command                                          | Description                               |
|--------------------------------------------------|-------------------------------------------|
| <code>docker compose up</code>                   | Create and start containers               |
| <code>docker compose up -d</code>                | Create and start containers in background |
| <code>docker compose up --build</code>           | Rebuild images before starting containers |
| <code>docker compose stop</code>                 | Stop services                             |
| <code>docker compose down</code>                 | Stop and remove containers and networks   |
| <code>docker compose ps</code>                   | List running containers                   |
| <code>docker compose logs</code>                 | View the logs of all containers           |
| <code>docker compose logs &lt;service&gt;</code> | View the logs of a specific service       |
| <code>docker compose logs -f</code>              | View and follow the logs                  |
| <code>docker compose pull</code>                 | Pull the latest images                    |
| <code>docker compose build</code>                | Build or rebuild services                 |
| <code>docker compose build --pull</code>         | Pull latest images before building        |

## Dockerfile instructions

| Instruction                      | Description                                     |
|----------------------------------|-------------------------------------------------|
| FROM <image>                     | Set the base image                              |
| FROM <image> AS <name>           | Set the base image and name the build stage     |
| RUN <command>                    | Execute a command as part of the build process  |
| RUN ["exec", "param1", "param2"] | Execute a command as part of the build process  |
| CMD ["exec", "param1", "param2"] | Execute a command when the container starts     |
| ENTRYPOINT ["exec", "param1"]    | Configure the container to run as an executable |
| ENV <key>=<value>                | Set an environment variable                     |
| EXPOSE <port>                    | Expose a port                                   |
| COPY <src> <dest>                | Copy files from source to destination           |
| COPY --from=<name> <src> <dest>  | Copy files from a build stage to destination    |
| WORKDIR <path>                   | Set the working directory                       |
| VOLUME <path>                    | Create a mount point                            |
| USER <user>                      | Set the user                                    |
| ARG <name>                       | Define a build argument                         |
| ARG <name>=<default>             | Define a build argument with a default value    |
| LABEL <key>=<value>              | Set a metadata label                            |
| HEALTHCHECK <command>            | Set a healthcheck command                       |

See <https://docs.docker.com/engine/reference/builder/> for the full Dockerfile reference.

## Example compose.yaml

Note: this file used to be called docker-compose.yaml, but now compose.yaml is preferred.

```
services:
  service1:
    image: <image>
    build: .
    volumes:
      - ./code:ro
    ports:
      - "8000:80"
    environment:
      KEY: value
```

## Docker Compose file reference

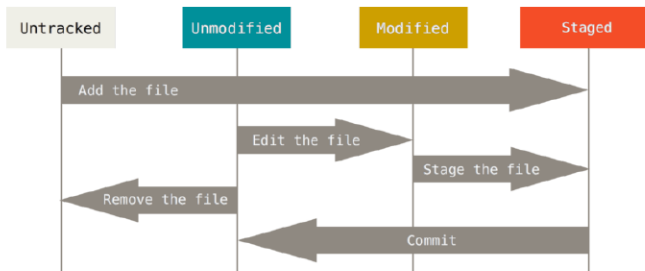
| Key                              | Description                                          |
|----------------------------------|------------------------------------------------------|
| name                             | Set the name of the project                          |
| services                         | A list of services defined in the file               |
| services.<name>.image            | Set the image to use or build                        |
| services.<name>.build            | Build context and options                            |
| services.<name>.build.context    | Build context (default is the current directory)     |
| services.<name>.build.dockerfile | Dockerfile to use (default is Dockerfile)            |
| services.<name>.build.target     | Build stage to use                                   |
| services.<name>.build.args       | Build arguments                                      |
| services.<name>.command          | Override the default command for the container       |
| services.<name>.entrypoint       | Override the default entrypoint for the container    |
| services.<name>.volumes          | Mount volumes in the container                       |
| services.<name>.ports            | Publish container ports to the host                  |
| services.<name>.environment      | Set environment variables in the container           |
| services.<name>.restart          | Restart policy (no/always/on-failure/unless-stopped) |
| services.<name>.scale            | Set the number of containers to run                  |
| services.<name>.networks         | List of networks to connect the container to         |
| services.<name>.depends_on       | List of services to start before this service        |
| services.<name>.labels           | Set metadata labels for the container                |
| networks                         | A list of networks defined in the file               |
| networks.<name>.driver           | Set the network driver                               |
| networks.<name>.external         | Don't create the network, use an existing one        |
| volumes                          | A list of volumes defined in the file                |
| volumes.<name>.name              | Set the name of the volume                           |
| volumes.<name>.driver            | Set the volume driver                                |
| configs                          | A list of configs defined in the file                |
| secrets                          | A list of secrets defined in the file                |

See <https://docs.docker.com/compose/compose-file/> for the full Docker Compose file reference.



# Git Cheat Sheet

## Workflow



## Start a new repository

| Command                                          | Description                                      |
|--------------------------------------------------|--------------------------------------------------|
| <code>git init</code>                            | Create a new repository in the current directory |
| <code>git clone &lt;url&gt; [&lt;dir&gt;]</code> | Clone a remote repository                        |

## History

| Command                             | Description                               |
|-------------------------------------|-------------------------------------------|
| <code>git log</code>                | Show commit logs                          |
| <code>git log --oneline</code>      | Show commit logs with one line per commit |
| <code>git show</code>               | Show the changes of the last commit       |
| <code>git shortlog</code>           | Summarize git log output by author        |
| <code>git blame &lt;file&gt;</code> | Show who changed which line in a file     |

## Local changes

| Command                           | Description                                                     |
|-----------------------------------|-----------------------------------------------------------------|
| <code>git status</code>           | List which files are staged, unstaged, and untracked            |
| <code>git add &lt;file&gt;</code> | Add a file to the staging area                                  |
| <code>git add -p</code>           | Interactively choose hunks of a file to add to the staging area |

## Local changes (cont.)

| Command                                    | Description                                                       |
|--------------------------------------------|-------------------------------------------------------------------|
| <code>git diff</code>                      | Show unstaged changes                                             |
| <code>git diff --staged</code>             | Show staged changes                                               |
| <code>git commit</code>                    | Commit staged changes using the default editor                    |
| <code>git commit -m &lt;message&gt;</code> | Commit staged changes with a message provided on the command line |
| <code>git commit --amend</code>            | Amend the last commit                                             |
| <code>git reset HEAD &lt;file&gt;</code>   | Unstage a file                                                    |
| <code>git checkout -- &lt;file&gt;</code>  | Discard local changes in a file                                   |

## Branches

| Command                                     | Description                           |
|---------------------------------------------|---------------------------------------|
| <code>git branch</code>                     | List local branches                   |
| <code>git branch -r</code>                  | List remote branches                  |
| <code>git branch &lt;branch&gt;</code>      | Create a new branch                   |
| <code>git checkout &lt;branch&gt;</code>    | Switch to a branch                    |
| <code>git checkout -b &lt;branch&gt;</code> | Create a new branch and switch to it  |
| <code>git merge &lt;branch&gt;</code>       | Merge a branch into the active branch |
| <code>git branch -d &lt;branch&gt;</code>   | Delete a branch                       |

## Tags

| Command                             | Description             |
|-------------------------------------|-------------------------|
| <code>git tag</code>                | List tags               |
| <code>git tag &lt;tag&gt;</code>    | Create a tag            |
| <code>git tag -a &lt;tag&gt;</code> | Create an annotated tag |
| <code>git tag -s &lt;tag&gt;</code> | Create a GPG-signed tag |
| <code>git tag -d &lt;tag&gt;</code> | Delete a tag            |
| <code>git show &lt;tag&gt;</code>   | Show the tag data       |

## Collaboration

| Command                                                      | Description                                                            |
|--------------------------------------------------------------|------------------------------------------------------------------------|
| <code>git remote -v</code>                                   | List remote repositories                                               |
| <code>git remote add &lt;name&gt; &lt;url&gt;</code>         | Add a new remote repository                                            |
| <code>git fetch &lt;remote&gt;</code>                        | Download all changes from a remote repository but don't integrate them |
| <code>git pull &lt;remote&gt;</code>                         | Download changes and directly merge/integrate them                     |
| <code>git pull &lt;remote&gt; &lt;branch&gt;</code>          | Download changes and directly merge/integrate a remote branch          |
| <code>git push &lt;remote&gt; &lt;branch&gt;</code>          | Upload local changes to a remote repository                            |
| <code>git push &lt;remote&gt; --delete &lt;branch&gt;</code> | Delete a remote branch                                                 |
| <code>git push &lt;remote&gt; --tags</code>                  | Push tags to a remote repository                                       |

## Stashing

| Command                      | Description                                         |
|------------------------------|-----------------------------------------------------|
| <code>git stash</code>       | Stash the changes in a dirty working directory away |
| <code>git stash list</code>  | List all stashes                                    |
| <code>git stash pop</code>   | Apply the last stash and delete it                  |
| <code>git stash apply</code> | Apply the last stash but don't delete it            |
| <code>git stash drop</code>  | Delete the last stash                               |
| <code>git stash clear</code> | Delete all stashes                                  |

## Writing good commit messages

1. Separate subject from body with a blank line
2. Limit the subject line to 50 characters
3. Capitalize the subject line
4. Do not end the subject line with a period
5. Use the imperative mood in the subject line
6. Wrap the body at 72 characters
7. Use the body to explain what and why vs. how

# Helm Cheat Sheet



## Helm

Helm is the package manager for Kubernetes. See <https://helm.sh/docs/intro/install/> for installation instructions.

## Global flags

| Flag                                     | Description                           |
|------------------------------------------|---------------------------------------|
| <code>--kube-context &lt;name&gt;</code> | Name of the Kubernetes context to use |
| <code>--namespace &lt;name&gt;</code>    | Namespace to use for this operation   |

## Repository management

| Command                                             | Description                                |
|-----------------------------------------------------|--------------------------------------------|
| <code>helm repo add &lt;name&gt; &lt;url&gt;</code> | Add a repository                           |
| <code>helm repo list</code>                         | List all added repositories                |
| <code>helm repo update</code>                       | Update the local cache of available charts |
| <code>helm repo remove &lt;name&gt;</code>          | Remove a repository                        |
| <code>helm search repo</code>                       | List all charts in the repositories        |
| <code>helm search repo &lt;keyword&gt;</code>       | Search for a chart in the repositories     |

## Installing Helm charts

| Command                                                                              | Description                                   |
|--------------------------------------------------------------------------------------|-----------------------------------------------|
| <code>helm install &lt;name&gt; &lt;chart&gt;</code>                                 | Install a chart with a name                   |
| <code>helm install &lt;chart&gt; --generate-name</code>                              | Install a chart, auto-generating a name       |
| <code>helm install &lt;name&gt; &lt;chart&gt; --namespace &lt;namespace&gt;</code>   | Install a chart in a specific namespace       |
| <code>helm install &lt;name&gt; &lt;chart&gt; --set &lt;key&gt;=&lt;value&gt;</code> | Install a chart with specific values          |
| <code>helm install &lt;name&gt; &lt;chart&gt; --values &lt;file&gt;</code>           | Install a chart using a values file           |
| <code>helm install &lt;name&gt; &lt;chart&gt; --dry-run --debug</code>               | Run a test installation to validate the chart |
| <code>helm install &lt;name&gt; &lt;chart&gt; --verify</code>                        | Verify the package before installing          |
| <code>helm install &lt;name&gt; &lt;chart&gt; --dependency-update</code>             | Update dependencies before installing         |
| <code>helm uninstall &lt;name&gt;</code>                                             | Uninstall a release                           |
| <code>helm uninstall &lt;name&gt; --keep-history</code>                              | Uninstall a release, keeping the history      |

## Listing releases

| Command                                               | Description                                                  |
|-------------------------------------------------------|--------------------------------------------------------------|
| <code>helm list</code>                                | List all releases in the current namespace                   |
| <code>helm list --all-namespaces</code>               | List all releases in all namespaces                          |
| <code>helm list -l &lt;label&gt;=&lt;value&gt;</code> | List releases with a specific label                          |
| <code>helm list --date</code>                         | List releases sorted by date                                 |
| <code>helm list --(pending failed uninstalled)</code> | List releases that are in a pending/failed/uninstalled state |
| <code>helm status &lt;name&gt;</code>                 | Show the status of a release                                 |

## Managing releases

| Command                                                                              | Description                               |
|--------------------------------------------------------------------------------------|-------------------------------------------|
| <code>helm upgrade &lt;name&gt; &lt;chart&gt;</code>                                 | Upgrade a release                         |
| <code>helm upgrade &lt;name&gt; &lt;chart&gt; --atomic</code>                        | Upgrade a release atomically              |
| <code>helm upgrade &lt;name&gt; &lt;chart&gt; --dependency-update</code>             | Upgrade a release and update dependencies |
| <code>helm upgrade &lt;name&gt; &lt;chart&gt; --version &lt;version&gt;</code>       | Upgrade a release to a specific version   |
| <code>helm upgrade &lt;name&gt; &lt;chart&gt; --set &lt;key&gt;=&lt;value&gt;</code> | Upgrade a release with specific values    |
| <code>helm rollback &lt;release&gt; &lt;revision&gt;</code>                          | Rollback a release to a previous revision |

## Developing charts

| Command                                                                               | Description                                  |
|---------------------------------------------------------------------------------------|----------------------------------------------|
| <code>helm create &lt;name&gt;</code>                                                 | Create a new chart                           |
| <code>helm package &lt;chart-path&gt;</code>                                          | Package a chart directory into a chart file  |
| <code>helm lint &lt;chart&gt;</code>                                                  | Lint a chart                                 |
| <code>helm show all &lt;chart&gt;</code>                                              | Inspect a chart and list all resources       |
| <code>helm show values &lt;chart&gt;</code>                                           | Inspect a chart and show default values      |
| <code>helm template &lt;name&gt; &lt;chart&gt;</code>                                 | Render templates locally                     |
| <code>helm template &lt;name&gt; &lt;chart&gt; --set &lt;key&gt;=&lt;value&gt;</code> | Render templates locally and override values |

# HTTP Cheat Sheet

## Example Request & Response

```
GET /hello.txt HTTP/1.1
Host: www.example.com
User-Agent: Mozilla/5.0
Accept-Language: en, nl
```

```
HTTP/1.1 200 OK
Date: Mon, 27 Jul 2020 12:28:53 GMT
Server: Apache/2.2.22
Content-Length: 12
Content-Type: text/plain
```

Hello World!

## Methods

| Method                     | Description                                                     |
|----------------------------|-----------------------------------------------------------------|
| GET (safe, idempotent)     | Transfer a current representation of the resource               |
| HEAD (safe, idempotent)    | Same as GET, but without transferring content                   |
| POST                       | Process the content of the request                              |
| PUT (idempotent)           | Update the resource with the content of the request             |
| DELETE (idempotent)        | Remove all current representations of the resource              |
| CONNECT                    | Establish a tunnel to the server                                |
| OPTIONS (safe, idempotent) | Describe the communication options for the target resource      |
| TRACE (safe, idempotent)   | Perform a message loop-back test along the path to the resource |

"Safe" methods are read-only and should not change the server state. "Idempotent" methods can be called multiple times without different outcomes. See [RFC 9110](#) for detailed information about each method.

## Request Headers

| Header                           | Description                                             |
|----------------------------------|---------------------------------------------------------|
| Referer: <url>                   | The referring web page URL                              |
| User-Agent: <ua>                 | Information about the client making the request         |
| Authorization: <scheme> <params> | Authenticate using the specified scheme and credentials |
| Accept: <media-types>            | The media types that the client can understand          |
| Accept-Encoding: <algorithms>    | The encoding algorithms the client understands          |
| Accept-Language: <langs>         | The languages that the client can understand            |

## Response Headers

| Header                      | Description                                    |
|-----------------------------|------------------------------------------------|
| Location: <url>             | The URL of the new resource or redirect target |
| Allow: <methods>            | The methods supported by the resource          |
| Server: <name>              | Information about the server software          |
| Retry-After: <s>            | Time to wait before a new request              |
| Age: <s>                    | The age of the cached response (in sec)        |
| Cache-Control: <directives> | Directives for caching mechanisms              |
| Expires: <date>             | Date/time after which the response expires     |

See [RFC 9111](#) for detailed information about caching.

## Status Codes

A detailed list of status codes can be found in [RFC 9110](#).

## Status Codes: Informational

| Code                    | Description                                                                              |
|-------------------------|------------------------------------------------------------------------------------------|
| 100 Continue            | The server has received the request headers and the client should proceed                |
| 101 Switching Protocols | The server will switch protocols to those defined by the response's Upgrade header field |

## Status Codes: Success

| Code                              | Description                                                    |
|-----------------------------------|----------------------------------------------------------------|
| 200 OK                            | The request has succeeded                                      |
| 201 Created                       | A new resource has been created                                |
| 202 Accepted                      | The request was accepted but processing has not been completed |
| 203 Non-Authoritative Information | The response content has been modified by a proxy              |
| 204 No Content                    | The request was fulfilled but no content is returned           |
| 205 Reset Content                 | Request fulfilled; the user agent should reset the view        |
| 206 Partial Content               | The server has fulfilled a partial GET request                 |

## Status Codes: Redirection

| Code                  | Description                                                               |
|-----------------------|---------------------------------------------------------------------------|
| 300 Multiple Choices  | The requested resource corresponds to any one of a set of representations |
| 301 Moved Permanently | The requested resource has been assigned a new permanent URI              |
| 302 Found             | The requested resource resides temporarily under a different URI          |



## Status Codes: Redirection (cont.)

| Code                   | Description                                                   |
|------------------------|---------------------------------------------------------------|
| 303 See Other          | A different resource can provide an indirect response         |
| 304 Not Modified       | The resource has not been modified since the last request     |
| 307 Temporary Redirect | Temporary redirect; client must not change the request method |
| 308 Permanent Redirect | Permanent redirect; client must not change the request method |

## Status Codes: Client Error

| Code                              | Description                                                            |
|-----------------------------------|------------------------------------------------------------------------|
| 400 Bad Request                   | The server cannot / will not process the request due to a client error |
| 401 Unauthorized                  | The request requires user authentication                               |
| 403 Forbidden                     | The server understood but refuses the request                          |
| 404 Not Found                     | No resource was found for the request URI                              |
| 405 Method Not Allowed            | The method is not allowed for the resource                             |
| 406 Not Acceptable                | No representation matches the Accept - * headers                       |
| 407 Proxy Authentication Required | The client must authenticate with the proxy server                     |
| 408 Request Timeout               | The server timed out waiting for the request                           |
| 409 Conflict                      | The request conflicts with the current resource state                  |
| 410 Gone                          | The requested resource is permanently unavailable                      |
| 411 Length Required               | The server requires a Content-Length header                            |
| 412 Precondition Failed           | One or more conditions evaluated to false                              |
| 413 Content Too Large             | The request content is too large                                       |
| 414 URI Too Long                  | The target URI was too long                                            |

## Status Codes: Client Error (cont.)

| Code                       | Description                                                                                  |
|----------------------------|----------------------------------------------------------------------------------------------|
| 415 Unsupported Media Type | The requested media type is not supported                                                    |
| 416 Range Not Satisfiable  | The request Range header is not satisfiable                                                  |
| 417 Expectation Failed     | The server cannot meet the Expect header                                                     |
| 421 Misdirected Request    | The request was directed at a server that is not able to produce a response for this request |
| 422 Unprocessable Content  | The server understands the content type but cannot process the content                       |
| 426 Upgrade Required       | The client should switch to a different protocol                                             |

## Status Codes: Server Error

| Code                           | Description                                                                   |
|--------------------------------|-------------------------------------------------------------------------------|
| 500 Internal Server Error      | The server encountered an unexpected error                                    |
| 501 Not Implemented            | The server does not support the functionality required to fulfill the request |
| 502 Bad Gateway                | The upstream server returned an invalid response                              |
| 503 Service Unavailable        | The server is currently unable to handle the request                          |
| 504 Gateway Timeout            | The upstream server did not respond in time                                   |
| 505 HTTP Version Not Supported | The server does not support the HTTP version                                  |

# Kubernetes Cheat Sheet

## Installation

Install the `kubectl` command line tool to interact with the Kubernetes API:

<https://kubernetes.io/docs/tasks/tools/#kubectl>

Enable autocompletion in bash:

```
composer completion bash | sudo tee /etc/bash_completion.d/kubectl
```

## Global flags

| Flag                                       | Description                            |
|--------------------------------------------|----------------------------------------|
| <code>--namespace &lt;namespace&gt;</code> | The name of the namespace to use       |
| <code>--context &lt;context&gt;</code>     | The name of the context to use         |
| <code>--help</code>                        | Show information about a given command |

## Context and configuration

| Command                                                    | Description                                      |
|------------------------------------------------------------|--------------------------------------------------|
| <code>kubectl config get-contexts</code>                   | List all contexts                                |
| <code>kubectl config current-context</code>                | Display the current context                      |
| <code>kubectl config use-context &lt;context&gt;</code>    | Switch to another context                        |
| <code>kubectl config delete-context &lt;context&gt;</code> | Delete the specified context from the kubeconfig |

## Display resources

| Command                                                                               | Description                                              |
|---------------------------------------------------------------------------------------|----------------------------------------------------------|
| <code>kubectl get &lt;resource&gt;</code>                                             | List all resources of this type in the current namespace |
| <code>kubectl get &lt;resource&gt; -o wide</code>                                     | List all resources with more details                     |
| <code>kubectl get &lt;resource&gt; -A</code>                                          | List all resources of this type in all namespaces        |
| <code>kubectl get &lt;resource&gt; &lt;name&gt;</code>                                | List a particular resource                               |
| <code>kubectl get &lt;resource&gt; &lt;name&gt; -o yaml</code>                        | Print a particular resource in YAML format               |
| <code>kubectl get &lt;resource&gt; &lt;name&gt; -l &lt;key1&gt;=&lt;value1&gt;</code> | List resources where label <key1> contains <value1>      |
| <code>kubectl describe &lt;resource&gt;</code>                                        | Show detailed information about a resource               |

## Apply configuration manifests

| Command                                    | Description                                |
|--------------------------------------------|--------------------------------------------|
| <code>kubectl apply -f &lt;file&gt;</code> | Apply a manifest from a file               |
| <code>kubectl apply -f &lt;dir&gt;</code>  | Apply all manifests in a directory         |
| <code>kubectl apply -k &lt;dir&gt;</code>  | Apply resources from a kustomize directory |

## Create resources manually

| Command                                                                                                                                                              | Description                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| <code>kubectl run &lt;name&gt; --image=&lt;image&gt;</code>                                                                                                          | Start a pod                                        |
| <code>kubectl create deployment &lt;name&gt; --image=&lt;image&gt;</code>                                                                                            | Create a deployment                                |
| <code>kubectl expose pod &lt;pod&gt; --port=&lt;port&gt;</code>                                                                                                      | Create a service for an existing pod               |
| <code>kubectl expose deployment &lt;name&gt; --port=&lt;port&gt;</code>                                                                                              | Create a service for an existing deployment        |
| <code>kubectl create ingress &lt;name&gt; --rule=&lt;host/path=svc:port&gt;</code>                                                                                   | Create an ingress that routes traffic to a service |
| <code>kubectl create job &lt;name&gt; --image=&lt;image&gt;</code>                                                                                                   | Create a job                                       |
| <code>kubectl create job &lt;name&gt; --from=cronjob/&lt;name&gt;</code>                                                                                             | Create a job from a cronjob                        |
| <code>kubectl create cronjob &lt;name&gt; --image=&lt;image&gt; --schedule=&lt;schedule&gt;</code>                                                                   | Create a cronjob, using a schedule in Cron format  |
| <code>kubectl create secret generic &lt;name&gt; --from-literal=&lt;key&gt;=&lt;value&gt;</code>                                                                     | Create a secret containing <key> and <value>       |
| <code>kubectl create secret docker-registry &lt;name&gt; --docker-server=&lt;server&gt; --docker-username=&lt;username&gt; --docker-password=&lt;password&gt;</code> | Create a secret for a Docker registry              |

## Generate YAML configuration manifests

| Command                                                                                            | Description                                  |
|----------------------------------------------------------------------------------------------------|----------------------------------------------|
| <code>kubectl create deployment &lt;name&gt; --image=&lt;image&gt; --dry-run=client -o yaml</code> | Generate a deployment manifest               |
| <code>kubectl expose deployment &lt;name&gt; --port=&lt;port&gt; --dry-run=client -o yaml</code>   | Generate a service manifest for a deployment |

## Edit resources

| Command                                                                                      | Description                              |
|----------------------------------------------------------------------------------------------|------------------------------------------|
| <code>kubectl edit &lt;resource&gt; &lt;name&gt;</code>                                      | Edit a resource in a text editor         |
| <code>kubectl set image &lt;resource&gt; &lt;name&gt; &lt;container&gt;=&lt;image&gt;</code> | Update the image of a container in a pod |

## Set labels and annotations

| Command                                                                               | Description                     |
|---------------------------------------------------------------------------------------|---------------------------------|
| <code>kubectl label &lt;resource&gt; &lt;name&gt; &lt;key&gt;=&lt;value&gt;</code>    | Add a label to a resource       |
| <code>kubectl annotate &lt;resource&gt; &lt;name&gt; &lt;key&gt;=&lt;value&gt;</code> | Add an annotation to a resource |

## Delete resources

| Command                                                   | Description                                                        |
|-----------------------------------------------------------|--------------------------------------------------------------------|
| <code>kubectl delete &lt;resource&gt; &lt;name&gt;</code> | Delete a particular resource                                       |
| <code>kubectl delete &lt;resource&gt; --all</code>        | Delete all resources of a particular type in the current namespace |
| <code>kubectl delete -f &lt;file&gt;</code>               | Delete a resource from a file                                      |

## Manage deployments

| Command                                                                                    | Description                                         |
|--------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <code>kubectl rollout status deployment &lt;name&gt;</code>                                | Show the status of a deployment rollout             |
| <code>kubectl rollout history deployment &lt;name&gt;</code>                               | View the rollout history of a deployment            |
| <code>kubectl rollout undo deployment &lt;name&gt;</code>                                  | Undo a previous rollout deployment                  |
| <code>kubectl rollout restart deployment &lt;name&gt;</code>                               | Restart a deployment                                |
| <code>kubectl scale deployment &lt;name&gt; --replicas=&lt;n&gt;</code>                    | Scale a deployment to <n> replicas                  |
| <code>kubectl autoscale deployment &lt;name&gt; --min=&lt;min&gt; --max=&lt;max&gt;</code> | Autoscale a deployment between <n> and <n> replicas |

## Execute commands

| Command                                                  | Description                        |
|----------------------------------------------------------|------------------------------------|
| <code>kubectl exec &lt;pod&gt; -- &lt;command&gt;</code> | Execute a command in a running pod |
| <code>kubectl exec -it &lt;pod&gt; -- sh</code>          | Open a shell in a running pod      |

## View logs

| Command                                  | Description                                 |
|------------------------------------------|---------------------------------------------|
| <code>kubectl logs &lt;pod&gt;</code>    | Print the logs for a pod                    |
| <code>kubectl logs -f &lt;pod&gt;</code> | Print the logs for a pod and keep streaming |

## Resource usage

| Command                       | Description                               |
|-------------------------------|-------------------------------------------|
| <code>kubectl top node</code> | Show resource (CPU/memory) usage of nodes |
| <code>kubectl top pod</code>  | Show resource (CPU/memory) usage of pods  |

## Other commands

| Command                            | Description                                     |
|------------------------------------|-------------------------------------------------|
| <code>kubectl version</code>       | Show the version of the client and server       |
| <code>kubectl api-resources</code> | Print the supported API resources on the server |

# Linux Cheat Sheet

## File Commands

| Command        | Description                                         |
|----------------|-----------------------------------------------------|
| cat <file>     | Print file contents                                 |
| touch <file>   | Create file (or update timestamp)                   |
| cp <src> <dst> | Copy a file                                         |
| mv <src> <dst> | Move / rename a file                                |
| rm <file>      | Remove file                                         |
| head <file>    | Print first 10 lines of file                        |
| tail <file>    | Print last 10 lines of file                         |
| tail -f <file> | Print last 10 lines and any new lines added to file |

## Directory Commands

| Command           | Description                                         |
|-------------------|-----------------------------------------------------|
| ls                | List files in the current directory                 |
| ls <dir>          | List files in <dir>                                 |
| ls -l             | List files with details                             |
| ls -a             | List all files (including hidden)                   |
| ls -al            | List all files with details                         |
| pwd               | Print the current/working directory                 |
| cd <dir>          | Change current directory to <dir>                   |
| cd ..             | Go up one directory                                 |
| mkdir <dir>       | Create directory                                    |
| mkdir -p <dir>    | Create directory and any missing parent directories |
| cp -r <src> <dst> | Copy a directory and its contents                   |
| mv <src> <dst>    | Move / rename a directory                           |
| rmdir <dir>       | Remove an empty directory                           |
| rm -r <dir>       | Remove a directory and its contents                 |

## Directory Aliases

| Alias | Directory         |
|-------|-------------------|
| .     | Current directory |
| ..    | Parent directory  |
| ~     | Home directory    |

## Link Commands

| Command               | Description                 |
|-----------------------|-----------------------------|
| ln -s <target> <link> | Create symbolic link        |
| ln <target> <link>    | Create hard link            |
| readlink -f <file>    | Print absolute path of file |

## Search Commands

| Command                  | Description                                        |
|--------------------------|----------------------------------------------------|
| grep <pattern> <file...> | Search for pattern in file(s)                      |
| grep -r <pattern> <dir>  | Search for pattern recursively in a directory      |
| grep -i <pattern> <file> | Search for pattern case-insensitively              |
| grep -v <pattern> <file> | Search for lines not matching pattern              |
| grep -n <pattern> <file> | Search and print line numbers                      |
| find <dir> -name <name>* | Find files in <dir> with name starting with <name> |
| find <dir> -type <type>  | Find files in <dir> of type <type>                 |

## Superuser Commands

| Command    | Description                     |
|------------|---------------------------------|
| sudo <cmd> | Run command as superuser (root) |
| su         | Switch to superuser (root)      |
| su <user>  | Switch to user <user>           |

## File Permission Commands

| Command                      | Description                                                                          |
|------------------------------|--------------------------------------------------------------------------------------|
| chmod <mode> <file>          | Change file permissions                                                              |
| chmod u+x <file>             | Add execute permission for the owner                                                 |
| chmod 644 <file>             | Set file permissions to 644 (read/write for owner, read for group and everyone else) |
| chmod -R g+w <dir>           | Add write permission for the group recursively to a directory                        |
| chown <owner> <file>         | Change file owner                                                                    |
| chown -R <owner> <dir>       | Change directory owner recursively                                                   |
| chown <owner>:<group> <file> | Change file owner and group                                                          |
| chgrp <group> <file>         | Change file group                                                                    |

## File Permission Bits

| Bit | Description |
|-----|-------------|
| 4   | Read (r)    |
| 2   | Write (w)   |
| 1   | Execute (x) |

First digit represents the owner, the second digit represents the group, and the third digit represents everyone else.

For example, 640 means read/write for owner, read for group, and no permissions for everyone else.

## File Permission User Types

| User Type | Description   |
|-----------|---------------|
| u         | Owner         |
| g         | Group         |
| o         | Everyone else |
| a         | All           |

## System Information Commands

| Command  | Description                   |
|----------|-------------------------------|
| df       | Show file system disk usage   |
| df -h    | Show disk usage in MB/GB/TB   |
| du       | Show disk usage per directory |
| du -h    | Show disk usage in MB/GB/TB   |
| free     | Show memory usage             |
| free -h  | Show memory usage in MB/GB/TB |
| uptime   | Show uptime                   |
| whoami   | Show current user             |
| uname    | Show system information       |
| uname -a | Show all system information   |

## Shutdown/Reboot Commands

| Command        | Description                         |
|----------------|-------------------------------------|
| shutdown       | Power off the system after 1 minute |
| shutdown now   | Power off the system immediately    |
| shutdown 16:30 | Power off the system at 16:30       |
| shutdown -c    | Cancel a scheduled shutdown         |
| reboot         | Reboot system                       |
| poweroff       | Power off system                    |

## Process Commands

| Command        | Description                    |
|----------------|--------------------------------|
| ps             | List processes                 |
| ps -ef         | List all processes             |
| top            | Display processes in real time |
| kill <pid>     | Kill process                   |
| pkill <name>   | Kill process with name         |
| killall <name> | Kill all processes with name   |

## Secure Shell (SSH) Commands

| Command                   | Description                                 |
|---------------------------|---------------------------------------------|
| ssh-keygen                | Generate SSH key pair                       |
| ssh-add <key>             | Add SSH key to SSH agent                    |
| ssh-add -l                | List SSH keys in SSH agent                  |
| ssh-copy-id <user>@<host> | Copy SSH key to <host> for <user>           |
| ssh <host>                | Connect to <host> as current user           |
| ssh <user>@<host>         | Connect to <host> as <user>                 |
| scp <src> <host>:<dest>   | Copy local file <src> to <dest> on <host>   |
| scp <host>:<src> <dest>   | Copy file <src> from <host> to local <dest> |

## APT Commands (Debian/Ubuntu)

| Command           | Description                     |
|-------------------|---------------------------------|
| apt install <pkg> | Install package                 |
| apt remove <pkg>  | Remove package                  |
| apt purge <pkg>   | Remove package and config files |
| apt update        | Update package list             |
| apt upgrade       | Upgrade packages                |
| apt search <term> | Search for package              |

## Other Commands

| Command       | Description                                      |
|---------------|--------------------------------------------------|
| echo <msg>    | Print message to stdout                          |
| man <cmd>     | Display manual page for <cmd>                    |
| history       | Display history of given commands                |
| which <cmd>   | Find path to executable <cmd>                    |
| whereis <cmd> | Locate binary, source, and manual page for <cmd> |

## IO Redirection

| Command         | Description                          |
|-----------------|--------------------------------------|
| <cmd> > <file>  | Replace contents of file with output |
| <cmd> >> <file> | Append output to file                |
| <cmd> < <file>  | Redirect input from file             |
| <cmd> 2> <file> | Redirect stderr to file              |
| <cmd> &> <file> | Redirect stdout and stderr to file   |

## Pipe Commands

| Command          | Description             |
|------------------|-------------------------|
| <cmd1>   <cmd2>  | Pipe stdout to next cmd |
| <cmd1>  & <cmd2> | Pipe stderr to next cmd |

## Bash Shortcuts

| Shortcut | Description             |
|----------|-------------------------|
| Ctrl + C | Stop current command    |
| Ctrl + A | Go to beginning of line |
| Ctrl + E | Go to end of line       |
| Ctrl + K | Cut current line        |
| Ctrl + L | Clear screen            |

## Directory Structure

| Directory | Description                      |
|-----------|----------------------------------|
| /         | Root                             |
| /bin      | Binary or executable programs    |
| /etc      | System configuration files       |
| /home     | User home directories            |
| /opt      | Optional or third-party software |
| /tmp      | Temporary files                  |
| /usr      | User programs                    |
| /var      | Variable data                    |
| /var/log  | Log files                        |



# Terraform Cheat Sheet



## Installation

See <https://developer.hashicorp.com/terraform/downloads> for installation instructions for your platform. Then run `terraform -install-autocomplete` to enable shell autocompletion.

## Initializing Terraform

| Command                              | Description                                              |
|--------------------------------------|----------------------------------------------------------|
| <code>terraform init</code>          | Prepare your working directory for other commands        |
| <code>terraform init -upgrade</code> | Upgrade modules/providers to the latest allowed versions |
| <code>terraform get</code>           | Only download and install modules                        |

## Making infrastructure changes

| Command                                                | Description                                            |
|--------------------------------------------------------|--------------------------------------------------------|
| <code>terraform plan</code>                            | Show changes required by the current configuration     |
| <code>terraform plan -out=&lt;file&gt;</code>          | Write the plan to a file to apply it later             |
| <code>terraform plan -target &lt;resource&gt;</code>   | Create a plan for a specific module or resource        |
| <code>terraform plan -replace &lt;resource&gt;</code>  | Force the plan to replace a specific resource          |
| <code>terraform plan -var 'key=value'</code>           | Set a value for one of the input variables             |
| <code>terraform plan -refresh-only</code>              | Inspect resource drift without updating the state file |
| <code>terraform apply</code>                           | Create or update infrastructure                        |
| <code>terraform apply &lt;file&gt;</code>              | Create or update infrastructure using a plan file      |
| <code>terraform apply -target &lt;resource&gt;</code>  | Create or update a specific resource                   |
| <code>terraform apply -replace &lt;resource&gt;</code> | Force the replacement of a specific resource           |
| <code>terraform apply -auto-approve</code>             | Skip interactive approval of plan before applying      |

## Inspecting output values

| Command                                         | Description                                 |
|-------------------------------------------------|---------------------------------------------|
| <code>terraform output</code>                   | Show all output values                      |
| <code>terraform output -json</code>             | Show all output values in JSON format       |
| <code>terraform output &lt;name&gt;</code>      | Show a specific output value                |
| <code>terraform output -raw &lt;name&gt;</code> | Show a specific output value without quotes |

## Managing state

| Command                                                               | Description                                   |
|-----------------------------------------------------------------------|-----------------------------------------------|
| <code>terraform show</code>                                           | Show the current state in human-readable form |
| <code>terraform show &lt;file&gt;</code>                              | Show a human-readable state or plan file      |
| <code>terraform show -json &lt;file&gt;</code>                        | Show a state or plan file in JSON format      |
| <code>terraform state list</code>                                     | List all resources in the state file          |
| <code>terraform state show &lt;resource&gt;</code>                    | Show details about a resource                 |
| <code>terraform state mv &lt;source&gt; &lt;dest&gt;</code>           | Rename a resource in the state file           |
| <code>terraform state rm &lt;resource&gt;</code>                      | Remove a resource from the state file         |
| <code>terraform state replace-provider &lt;from&gt; &lt;to&gt;</code> | Replace provider for resources in the state   |
| <code>terraform import &lt;resource&gt; &lt;remote_id&gt;</code>      | Import existing infrastructure into Terraform |
| <code>terraform state pull</code>                                     | Pull current state and output to stdout       |
| <code>terraform refresh</code>                                        | (Deprecated) Update state to match reality    |

## Tainting resources

| Command                                         | Description                                 |
|-------------------------------------------------|---------------------------------------------|
| <code>terraform taint &lt;resource&gt;</code>   | (Deprecated) Mark a resource to be replaced |
| <code>terraform untaint &lt;resource&gt;</code> | Mark a resource as no longer tainted        |

## Destroying infrastructure

| Command                                                 | Description                                 |
|---------------------------------------------------------|---------------------------------------------|
| <code>terraform destroy</code>                          | Destroy infrastructure managed by Terraform |
| <code>terraform destroy -target &lt;resource&gt;</code> | Destroy a specific resource                 |

## Formatting and validation

| Command                           | Description                                                                              |
|-----------------------------------|------------------------------------------------------------------------------------------|
| <code>terraform validate</code>   | Check whether the configuration is valid                                                 |
| <code>terraform fmt</code>        | Reformat your configuration in the standard style                                        |
| <code>terraform fmt -check</code> | Check whether the configuration is formatted correctly, return non-zero exit code if not |

## Terraform Cloud / Remote Authentication

| Command                     | Description                 |
|-----------------------------|-----------------------------|
| terraform login             | Log in to Terraform Cloud   |
| terraform login <hostname>  | Log in to a different host  |
| terraform logout            | Log out of Terraform Cloud  |
| terraform logout <hostname> | Log out of a different host |

## Managing workspaces

| Command                           | Description                            |
|-----------------------------------|----------------------------------------|
| terraform workspace list          | List all existing workspaces           |
| terraform workspace show          | Show the name of the current workspace |
| terraform workspace select <name> | Select a different workspace           |
| terraform workspace new <name>    | Create a new workspace                 |
| terraform workspace delete <name> | Delete an existing workspace           |

## Other commands

| Command                                 | Description                                        |
|-----------------------------------------|----------------------------------------------------|
| terraform providers                     | Show the providers required for this configuration |
| terraform force-unlock <lock-id>        | Release a stuck lock                               |
| terraform console                       | Try Terraform expressions at an interactive prompt |
| terraform graph   dot -Tpng > graph.png | Generate a visual graph of Terraform resources     |
| terraform version                       | Show the current Terraform version                 |
| terraform -help                         | Show help output for Terraform                     |
| terraform -help <command>               | Show help output for a specific Terraform command  |

## Referencing Named Values

| Syntax                  | Description                     |
|-------------------------|---------------------------------|
| <RESOURCE_TYPE>.<NAME>  | Reference to a managed resource |
| var.<NAME>              | Reference to an input variable  |
| local.<NAME>            | Reference to a local value      |
| module.<MODULE NAME>    | Reference to a child module     |
| data.<DATA TYPE>.<NAME> | Reference to a data source      |

## Conditional Expressions

| Syntax                   | Description                                               |
|--------------------------|-----------------------------------------------------------|
| condition ? true : false | If condition is true, return true, otherwise return false |

## Splat Expressions

| Syntax                                | Description                                                                    |
|---------------------------------------|--------------------------------------------------------------------------------|
| <RESOURCE_TYPE>.<NAME>[*].<ATTRIBUTE> | Return a list of values for the given attribute of all instances of a resource |

## Resource Meta-Arguments

| Argument   | Description                                                       |
|------------|-------------------------------------------------------------------|
| depends_on | Explicitly specify resource dependencies                          |
| count      | Create multiple instances of a resource                           |
| for_each   | Create an instance of a resource for each element in a map or set |
| provider   | Specify a provider configuration block to use for this resource   |
| lifecycle  | Configure the behavior of a resource over its lifetime            |

## Lifecycle Meta-Argument Attributes

| Argument              | Description                                           |
|-----------------------|-------------------------------------------------------|
| create_before_destroy | Create the new resource before destroying the old one |
| prevent_destroy       | Prevent Terraform from destroying the resource        |
| ignore_changes        | Ignore changes to specific resource attributes        |